

```
elad@Elad-Laptop: /mnt/c/Users/elad2/OneDrive/Desktop/terraform$ terraform plan
module.networking.data.aws_availability_zones.available: Reading...
module.networking.data.aws_availability_zones.available: Read complete after 1s [id-us-east-1]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# module.ec2.aws_instance.backend will be created
+ resource "aws_instance" "backend" {
  + ami                     = "ami-084568db4383264d4"
  + arn                    = (known after apply)
  + associate_public_ip_address = true
  + availability_zone       = (known after apply)
  + cpu_core_count          = (known after apply)
  + cpu_threads_per_core    = (known after apply)
  + disable_api_stop        = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized           = (known after apply)
  + enable_primary_ipv6     = (known after apply)
  + get_password_data       = false
  + host_id                 = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile    = "infra-automation-dev-backend-profile"
  + id                     = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle      = (known after apply)
  + instance_state          = (known after apply)
  + instance_type           = "t3.micro"
  + ipv6_address_count      = (known after apply)
  + ipv6_addresses         = (known after apply)
  + key_name               = "elad2"
  + monitoring             = (known after apply)
  + outpost_arn            = (known after apply)
  + password_data          = (known after apply)
  + placement_group        = (known after apply)
  + placement_partition_number = (known after apply)
  + primary_network_interface_id = (known after apply)
  + private_dns            = (known after apply)
  + private_ip             = (known after apply)
  + public_dns             = (known after apply)
  + public_ip              = (known after apply)
  + secondary_private_ips   = (known after apply)
  + security_groups        = (known after apply)
  + source_dest_check       = true
  + spot_instance_request_id = (known after apply)
  + subnet_id              = (known after apply)
  + tags                   = {
    + "Environment" = "dev"
    + "Name"        = "infra-automation-dev-backend"
    + "Role"        = "backend"
  }
+ tags_all                = {
  + "Environment" = "dev"
  + "Name"        = "infra-automation-dev-backend"
  + "Role"        = "backend"
}
+ tenancy                 = (known after apply)
+ user_data               = (known after apply)
+ user_data_base64        = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids  = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)
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+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)
}

# module.ec2.aws_instance.frontend will be created
+ resource "aws_instance" "frontend" {
  + ami                    = "ami-084568db4383264d4"
  + arn                    = (known after apply)
  + associate_public_ip_address = true
  + availability_zone       = (known after apply)
  + cpu_core_count          = (known after apply)
  + cpu_threads_per_core    = (known after apply)
  + disable_api_stop        = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized           = (known after apply)
  + enable_primary_ipv6     = (known after apply)
  + get_password_data       = false
  + host_id                 = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile    = (known after apply)
  + id                      = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle      = (known after apply)
  + instance_state          = (known after apply)
  + instance_type           = "t3.micro"
  + ipv6_address_count      = (known after apply)
  + ipv6_addresses          = (known after apply)
  + key_name                = "elad2"
  + monitoring              = (known after apply)
  + outpost_arn             = (known after apply)
  + password_data           = (known after apply)
  + placement_group         = (known after apply)
  + placement_partition_number = (known after apply)
  + primary_network_interface_id = (known after apply)
  + private_dns             = (known after apply)
  + private_ip              = (known after apply)
  + public_dns              = (known after apply)
  + public_ip               = (known after apply)
  + secondary_private_ips   = (known after apply)
  + security_groups         = (known after apply)
  + source_dest_check       = true
  + spot_instance_request_id = (known after apply)
  + subnet_id              = (known after apply)
  + tags                    = {
    + "Environment" = "dev"
    + "Name"        = "infra-automation-dev-frontend"
    + "Role"        = "frontend"
  }
  + tags_all = {
    + "Environment" = "dev"
    + "Name"        = "infra-automation-dev-frontend"
    + "Role"        = "frontend"
  }
  + tenancy = (known after apply)
  + user_data = (known after apply)
  + user_data_base64 = (known after apply)
  + user_data_replace_on_change = false
  + vpc_security_group_ids = (known after apply)

+ capacity_reservation_specification (known after apply)

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+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)
}

# module.ec2.aws_instance.worker will be created
+ resource "aws_instance" "worker" {
  + ami                    = "ami-084568db4383264d4"
  + arn                    = (known after apply)
  + associate_public_ip_address = true
  + availability_zone       = (known after apply)
  + cpu_core_count          = (known after apply)
  + cpu_threads_per_core    = (known after apply)
  + disable_api_stop        = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized           = (known after apply)
  + enable_primary_ipv6     = (known after apply)
  + get_password_data       = false
  + host_id                 = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile    = (known after apply)
  + id                     = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle      = (known after apply)
  + instance_state          = (known after apply)
  + instance_type           = "t3.micro"
  + ipv6_address_count      = (known after apply)
  + ipv6_addresses         = (known after apply)
  + key_name                = "elad2"
  + monitoring              = (known after apply)
  + outpost_arn             = (known after apply)
  + password_data           = (known after apply)
  + placement_group         = (known after apply)
  + placement_partition_number = (known after apply)
  + primary_network_interface_id = (known after apply)
  + private_dns             = (known after apply)
  + private_ip              = (known after apply)
  + public_dns              = (known after apply)
  + public_ip               = (known after apply)
  + secondary_private_ips   = (known after apply)
  + security_groups         = (known after apply)
  + source_dest_check       = true
  + spot_instance_request_id = (known after apply)
  + subnet_id               = (known after apply)
  + tags                    = {
    + "Environment" = "dev"
    + "Name"        = "infra-automation-dev-worker"
    + "Role"        = "worker"
  }
  + tags_all              = {
    + "Environment" = "dev"
    + "Name"        = "infra-automation-dev-worker"
    + "Role"        = "worker"
  }
  + tenancy                = (known after apply)

```

```

+ user_data_base64          = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids    = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)
}

# module.iam.aws_iam_instance_profile.backend_profile will be created
+ resource "aws_iam_instance_profile" "backend_profile" {
  + arn          = (known after apply)
  + create_date = (known after apply)
  + id          = (known after apply)
  + name        = "infra-automation-dev-backend-profile"
  + name_prefix = (known after apply)
  + path        = "/"
  + role        = "infra-automation-dev-backend-role"
  + tags_all    = (known after apply)
  + unique_id   = (known after apply)
}

# module.iam.aws_iam_policy.backend_policy will be created
+ resource "aws_iam_policy" "backend_policy" {
  + arn          = (known after apply)
  + attachment_count = (known after apply)
  + description   = "Allows backend EC2 to upload reports to S3 and publish SNS messages"
  + id          = (known after apply)
  + name        = "infra-automation-dev-backend-policy"
  + name_prefix = (known after apply)
  + path        = "/"
  + policy       = (known after apply)
  + policy_id    = (known after apply)
  + tags_all    = (known after apply)
}

# module.iam.aws_iam_role.backend_role will be created
+ resource "aws_iam_role" "backend_role" {
  + arn          = (known after apply)
  + assume_role_policy = jsonencode(
    {
      + Statement = [
        + {
          + Action   = "sts:AssumeRole"
          + Effect    = "Allow"
          + Principal = {
            + Service = "ec2.amazonaws.com"
          }
        }
      ],
    }
  )
  + Version = "2012-10-17"
}

+ create_date = (known after apply)

```

```

+ force_detach_policies = false
+ id                     = (known after apply)
+ managed_policy_arns   = (known after apply)
+ max_session_duration  = 3600
+ name                  = "infra-automation-dev-backend-role"
+ name_prefix           = (known after apply)
+ path                  = "/"
+ tags_all              = (known after apply)
+ unique_id             = (known after apply)

+ inline_policy (known after apply)
}

# module.iam.aws_iam_role_policy_attachment.backend_attach will be created
+ resource "aws_iam_role_policy_attachment" "backend_attach" {
+ id           = (known after apply)
+ policy_arn   = (known after apply)
+ role         = "infra-automation-dev-backend-role"
}

# module.networking.aws_internet_gateway.main will be created
+ resource "aws_internet_gateway" "main" {
+ arn          = (known after apply)
+ id           = (known after apply)
+ owner_id     = (known after apply)
+ tags         = {
+   + "Environment" = "dev"
+   + "Name"        = "infra-automation-dev-igw"
+ }
+ tags_all = {
+   + "Environment" = "dev"
+   + "Name"        = "infra-automation-dev-igw"
+ }
+ vpc_id       = (known after apply)
}

# module.networking.aws_route.internet_access will be created
+ resource "aws_route" "internet_access" {
+ destination_cidr_block = "0.0.0.0/0"
+ gateway_id            = (known after apply)
+ id                    = (known after apply)
+ instance_id           = (known after apply)
+ instance_owner_id     = (known after apply)
+ network_interface_id  = (known after apply)
+ origin                = (known after apply)
+ route_table_id        = (known after apply)
+ state                 = (known after apply)
}

# module.networking.aws_route_table.public will be created
+ resource "aws_route_table" "public" {
+ arn          = (known after apply)
+ id           = (known after apply)
+ owner_id     = (known after apply)
+ propagating_vgws = (known after apply)
+ route        = (known after apply)
+ tags         = {
+   + "Environment" = "dev"
+   + "Name"        = "infra-automation-dev-public-rt"
+ }
+ tags_all = {
+   + "Environment" = "dev"
+   + "Name"        = "infra-automation-dev-public-rt"
+ }
+ vpc_id       = (known after apply)
}

# module.networking.aws_route_table_association.public will be created
+ resource "aws_route_table_association" "public" {
+ id           = (known after apply)
+ route_table_id = (known after apply)
}

```



```
+ performance_insights_retention_period = (known after apply)
+ port                                 = 5432
+ publicly_accessible                  = false
+ replica_mode                         = (known after apply)
+ replicas                             = (known after apply)
+ resource_id                          = (known after apply)
+ skip_final_snapshot                  = true
+ snapshot_identifier                  = (known after apply)
+ status                               = (known after apply)
+ storage_throughput                   = (known after apply)
+ storage_type                         = (known after apply)
+ tags                                 = {
  + "Environment" = "dev"
  + "Name"         = "infra-automation-dev-postgres"
}
+ tags_all                               = {
  + "Environment" = "dev"
  + "Name"         = "infra-automation-dev-postgres"
}
+ timezone                               = (known after apply)
+ username                               = "postgres"
+ vpc_security_group_ids                 = (known after apply)
}
```

module.rds_postgresql.aws_db_subnet_group.main will be created

```
+ resource "aws_db_subnet_group" "main" {
  + arn                = (known after apply)
  + description         = "Managed by Terraform"
  + id                  = (known after apply)
  + name                = "infra-automation-dev-db-subnet-group"
  + name_prefix         = (known after apply)
  + subnet_ids          = (known after apply)
  + supported_network_types = (known after apply)
  + tags                = {
    + "Environment" = "dev"
    + "Name"         = "infra-automation-dev-db-subnet-group"
  }
  + tags_all            = {
    + "Environment" = "dev"
    + "Name"         = "infra-automation-dev-db-subnet-group"
  }
  + vpc_id              = (known after apply)
}
```

module.s3_bucket.aws_s3_bucket.main will be created

```
+ resource "aws_s3_bucket" "main" {
  + acceleration_status = (known after apply)
  + acl                  = (known after apply)
  + arn                  = (known after apply)
  + bucket                = "elad-devops-project-904053120094"
  + bucket_domain_name   = (known after apply)
  + bucket_prefix        = (known after apply)
  + bucket_regional_domain_name = (known after apply)
  + force_destroy        = false
  + hosted_zone_id       = (known after apply)
  + id                    = (known after apply)
  + object_lock_enabled  = (known after apply)
  + policy                = (known after apply)
  + region                = (known after apply)
  + request_payer        = (known after apply)
  + tags                 = {
    + "Environment" = "dev"
    + "Name"         = "elad-devops-project-904053120094"
  }
  + tags_all              = {
    + "Environment" = "dev"
    + "Name"         = "elad-devops-project-904053120094"
  }
  + website_domain        = (known after apply)
  + website_endpoint      = (known after apply)
}
```

```

+ cors_rule (known after apply)

+ grant (known after apply)

+ lifecycle_rule (known after apply)

+ logging (known after apply)

+ object_lock_configuration (known after apply)

+ replication_configuration (known after apply)

+ server_side_encryption_configuration (known after apply)

+ versioning (known after apply)

+ website (known after apply)
}

# module.s3_bucket.aws_s3_bucket_public_access_block.main will be created
+ resource "aws_s3_bucket_public_access_block" "main" {
  + block_public_acls      = true
  + block_public_policy    = true
  + bucket                = (known after apply)
  + id                    = (known after apply)
  + ignore_public_acls     = true
  + restrict_public_buckets = true
}

# module.security_groups.aws_security_group.backend_sg will be created
+ resource "aws_security_group" "backend_sg" {
  + arn                = (known after apply)
  + description        = "Security group for backend server"
  + egress              = [
    + {
      + cidr_blocks      = [
        + "0.0.0.0/0",
      ]
      + from_port        = 0
      + ipv6_cidr_blocks = []
      + prefix_list_ids  = []
      + protocol         = "-1"
      + security_groups  = []
      + self              = false
      + to_port          = 0
      # (1 unchanged attribute hidden)
    },
  ]
  + id                = (known after apply)
  + ingress            = [
    + {
      + cidr_blocks      = [
        + "62.56.148.113/32",
      ]
      + description      = "SSH access"
      + from_port        = 22
      + ipv6_cidr_blocks = []
      + prefix_list_ids  = []
      + protocol         = "tcp"
      + security_groups  = []
      + self              = false
      + to_port          = 22
    },
  ]
  + {
    + cidr_blocks      = []
    + description      = "Backend API from frontend"
    + from_port        = 5000
    + ipv6_cidr_blocks = []
    + prefix_list_ids  = []
    + protocol         = "tcp"
  }
}

```

```

+ security_groups = (known after apply)
+ self           = false
+ to_port        = 5000
},
]
+ name            = "infra-automation-dev-backend-sg"
+ name_prefix     = (known after apply)
+ owner_id        = (known after apply)
+ revoke_rules_on_delete = false
+ tags            = {
+   "Name" = "infra-automation-dev-backend-sg"
}
+ tags_all        = {
+   "Name" = "infra-automation-dev-backend-sg"
}
+ vpc_id          = (known after apply)
}

# module.security_groups.aws_security_group.frontend_sg will be created
+ resource "aws_security_group" "frontend_sg" {
+   arn            = (known after apply)
+   description     = "Security group for frontend server"
+   egress          = [
+     {
+       cidr_blocks = [
+         "0.0.0.0/0",
+       ]
+       from_port    = 0
+       ipv6_cidr_blocks = []
+       prefix_list_ids = []
+       protocol      = "-1"
+       security_groups = []
+       self          = false
+       to_port       = 0
+       # (1 unchanged attribute hidden)
+     },
+   ]
+   id              = (known after apply)
+   ingress          = [
+     {
+       cidr_blocks = [
+         "0.0.0.0/0",
+       ]
+       description = "HTTP access"
+       from_port    = 80
+       ipv6_cidr_blocks = []
+       prefix_list_ids = []
+       protocol      = "tcp"
+       security_groups = []
+       self          = false
+       to_port       = 80
+     },
+     {
+       cidr_blocks = [
+         "62.56.148.113/32",
+       ]
+       description = "SSH access"
+       from_port    = 22
+       ipv6_cidr_blocks = []
+       prefix_list_ids = []
+       protocol      = "tcp"
+       security_groups = []
+       self          = false
+       to_port       = 22
+     },
+   ],
+   name            = "infra-automation-dev-frontend-sg"
+   name_prefix     = (known after apply)
+   owner_id        = (known after apply)
+   revoke_rules_on_delete = false
+   tags            = {

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```

+ tags              = {
  + "Name" = "infra-automation-dev-frontend-sg"
}
+ tags_all          = {
  + "Name" = "infra-automation-dev-frontend-sg"
}
+ vpc_id            = (known after apply)
}

# module.security_groups.aws_security_group.rds_sg will be created
+ resource "aws_security_group" "rds_sg" {
  + arn              = (known after apply)
  + description      = "Security group for PostgreSQL RDS"
  + egress           = [
    + {
      + cidr_blocks = [
        + "0.0.0.0/0",
      ]
      + from_port   = 0
      + ipv6_cidr_blocks = []
      + prefix_list_ids = []
      + protocol    = "-1"
      + security_groups = []
      + self        = false
      + to_port     = 0
      # (1 unchanged attribute hidden)
    },
  ]
  + id              = (known after apply)
  + ingress         = [
    + {
      + cidr_blocks = []
      + description = "PostgreSQL from backend"
      + from_port   = 5432
      + ipv6_cidr_blocks = []
      + prefix_list_ids = []
      + protocol    = "tcp"
      + security_groups = (known after apply)
      + self        = false
      + to_port     = 5432
    },
  ]
  + name            = "infra-automation-dev-rds-sg"
  + name_prefix     = (known after apply)
  + owner_id        = (known after apply)
  + revoke_rules_on_delete = false
  + tags            = {
    + "Name" = "infra-automation-dev-rds-sg"
  }
  + tags_all        = {
    + "Name" = "infra-automation-dev-rds-sg"
  }
  + vpc_id          = (known after apply)
}

# module.security_groups.aws_security_group.worker_sg will be created
+ resource "aws_security_group" "worker_sg" {
  + arn              = (known after apply)
  + description      = "Security group for worker server"
  + egress           = [
    + {
      + cidr_blocks = [
        + "0.0.0.0/0",
      ]
      + from_port   = 0
      + ipv6_cidr_blocks = []
      + prefix_list_ids = []
      + protocol    = "-1"
      + security_groups = []
      + self        = false
      + to_port     = 0
    },
  ]

```

```

    ]
+ id              = (known after apply)
+ ingress         = [
+ {
+   + cidr_blocks      = [
+     + "62.56.148.113/32",
+   ]
+   + description      = "SSH access"
+   + from_port        = 22
+   + ipv6_cidr_blocks = []
+   + prefix_list_ids  = []
+   + protocol         = "tcp"
+   + security_groups  = []
+   + self             = false
+   + to_port          = 22
+ },
+ {
+   + cidr_blocks      = []
+   + description      = "Worker API from backend"
+   + from_port        = 5001
+   + ipv6_cidr_blocks = []
+   + prefix_list_ids  = []
+   + protocol         = "tcp"
+   + security_groups  = (known after apply)
+   + self             = false
+   + to_port          = 5001
+ },
+ ]
+ name            = "infra-automation-dev-worker-sg"
+ name_prefix     = (known after apply)
+ owner_id        = (known after apply)
+ revoke_rules_on_delete = false
+ tags            = {
+   + "Name" = "infra-automation-dev-worker-sg"
+ }
+ tags_all        = {
+   + "Name" = "infra-automation-dev-worker-sg"
+ }
+ vpc_id          = (known after apply)
}

```

module.sns_topic.aws_sns_topic.main will be created

```

+ resource "aws_sns_topic" "main" {
+   arn              = (known after apply)
+   beginning_archive_time = (known after apply)
+   content_based_deduplication = false
+   fifo_throughput_scope = (known after apply)
+   fifo_topic       = false
+   id               = (known after apply)
+   name             = "devops-prod-svc-topic"
+   name_prefix      = (known after apply)
+   owner            = (known after apply)
+   policy           = (known after apply)
+   signature_version = (known after apply)
+   tags_all         = (known after apply)
+   tracing_config   = (known after apply)
+ }

```

module.sns_topic.aws_sns_topic.subscription.email will be created

```

+ resource "aws_sns_topic_subscription" "email" {
+   arn              = (known after apply)
+   confirmation_timeout_in_minutes = 1
+   confirmation_was_authenticated = (known after apply)
+   endpoint         = "elad291297@gmail.com"
+   endpoint_auto_confirms = false
+   filter_policy_scope = (known after apply)
+   id               = (known after apply)
+   owner_id         = (known after apply)
+   pending_confirmation = (known after apply)
+   protocol         = "email"
+   raw_message_delivery = false
+   topic_arn        = (known after apply)
+ }

```

Plan: 25 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ backend_private_ip = (known after apply)
+ backend_public_ip  = (known after apply)
+ frontend_public_ip  = (known after apply)
+ rds_endpoint        = (known after apply)
+ s3_bucket_name      = "elad-devops-project-904053120094"
+ sns_topic_arn       = (known after apply)
+ worker_private_ip   = (known after apply)
+ worker_public_ip    = (known after apply)
```